A Resource for the Global Research Community

Contextual Considerations & Practical Guidelines for Direct Assessments of Children's Executive Function Skills

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INTRODUCTION

The main objective of this document is to foster greater research capacity to assess executive function (EF) skills in children ages 3-12 around the globe, and especially in low-resource settings.

These guidelines are general recommendations based on the **Global Executive Function Initiative's** (GEFI) members' collective experience conducting EF assessments in diverse cultures and contexts.

We compiled recommended practices for selection, adaptation, and administration of EF performance-based tasks and for processing, evaluating, reporting, and interpreting data that these tasks yield. While we derive these recommendations from our work using existing EF tasks, these recommendations can be used to guide the design and testing of novel EF tasks.

We hope this resource will help researchers conduct more psychometrically rigorous and ecologically valid assessments of children's EF skills, expanding scientific understanding of these skills across different cultures and contexts.

Although we considered a wide range of situations, this guide does not provide an exhaustive review of all issues. We acknowledge there is not always consensus about best practices. While we hope this resource eases the way for researchers who wish to incorporate and administer EF tasks into their studies, it is not a prescriptive blueprint. Researchers are ultimately responsible for making choices that best fit their specific study design, cultural context, and available resources.

Before using these guidelines, researchers should have a clear sense of:

1. **The rationale for assessing EF skills** (e.g., how direct assessment of EF skills fits within the study's research questions)
2. **The study population** (e.g., developmental stage, age range, language skills)
3. **Lived experiences and cultural values of the study population** that could affect how children experience standardized direct assessment procedures (e.g., formal schooling, interpersonal dynamics between children and adults who are not relatives)



Photo Credit: Bangladesh, Ishita Ahmed

SELECTING EXECUTIVE FUNCTION TASKS

There are Three Foundational Domains of EF Skills

1

INHIBITORY CONTROL

The ability to overcome or stop a habitual or over-learned response (i.e., response inhibition) and to maintain focused attention in the face of distraction or irrelevant information (i.e., interference suppression)

2

WORKING MEMORY

The ability to hold and manipulate verbal or nonverbal information in mind

3

COGNITIVE FLEXIBILITY

The ability to flexibly shift or switch between competing rules or demands



Photo Credit: South Africa, Project led by Cathi Draper

Construct Considerations

Given the effort and resources involved in conducting this type of research, when feasible, select multiple tasks for each domain that are relevant for the study aims. However, when EF skills are not the central focus of the study but merely a construct of interest (e.g., when EF is a secondary outcome in a program evaluation study), it may not be possible to include multiple tasks. In such cases, it is even more important to carefully select, adapt, and evaluate a couple of EF tasks that are best aligned with the study aims and the following considerations.

To help researchers with task selection, we summarize multiple paradigms that have been used to measure EF skills in global settings with children ages 3-12 (see Table 1). We include only paradigms for which task materials and apps are free and publicly available, and for which there is sufficient evidence of wide use across contexts to yield valid data. We emphasize the objectives of each paradigm, which should be considered when modifications to individual tasks are being considered. We also highlight key aspects of administration procedures to help researchers evaluate developmental and cultural appropriateness and list select countries where task variants have been used. Throughout this guide, we offer lessons learned, cautionary notes, and adaptation suggestions that may help researchers address deviations from these assumptions in a way that maintains validity of EF assessment. If the study context does not align with task assumptions, then we recommend exploring other tasks that are appropriate for the context.

TABLE 1

Recommended Executive Function Tasks

CONSTRUCT	PARADIGM DESCRIPTION	ADMINISTRATION PROCEDURE	EXEMPLAR STUDY SETTINGS
Inhibitory Control-Response Inhibition Non-Verbal	**Go/No-Go** Children are instructed to press a button when frequent "go" stimuli appear in order to build up a prepotent response and inhibit pressing a button when less frequent "no-go" stimuli appear. Without reliable "go" performance (e.g., high accuracy score), this task does not yield valid data.	**Common Stimuli:** animals, shapes, garden vegetables, emotions **Versions:** digital versions are more common. Manual versions have also been used with young children **Metrics:** digital version yields accuracy and response time data, manual version yields accuracy only	India, South Africa, Pakistan, Kenya, Brazil, Argentina, Ethiopia, Hungary
Inhibitory Control-Interference Suppression Verbal + Non-Verbal Variants	**Stroop** Children are instructed to say or select something that is the opposite (incongruent) of what is perceived (e.g., say/touch "night/moon" when shown a picture of the sun). Children need to overcome automatic tendencies to respond based on their perceptions and instead respond in accordance to the goals set forth in the instructions based on the different task variants. Children must be willing to say the opposite of what is presented to them for this task to be valid.	**Common stimuli:** sounds (select a picture of a dog when you hear meow), images (sun/moon), colors/words, numbers **Versions:** digital tablet-based versions or manual drawn pictures are used **Metrics:** accuracy data for both digital and manual; and response time for digital only	South Africa, Nigeria, Pakistan, Brazil, Kenya, Germany, Costa Rica, Cameroon, Madagascar
Inhibitory Control-Response Inhibition + Cognitive Flexibility Non-Verbal	**Simon Effect (Hearts and Flowers)** Children are presented with three different blocks of trials. In the first (congruent) block, children are instructed to press a button on the same side as the stimulus (heart) to build up a prepotent response. In the second (incongruent) block, children are instructed to press a button on the opposite side of a different stimulus (flower) to inhibit their prepotent response. In the third (mixed) block, children alternate between trials that present both stimuli (hearts and flowers) by switching between the different task rules. Performance on the second and third block are used to measure EF once children demonstrate proficiency (e.g., high accuracy score) in the first block.	**Common stimuli:** hearts and flowers, fruits, animals, spatial conflict arrows **Versions:** manual and digital tablet-based are used, digital is preferred **Metrics:** digital tablet-based versions yield accuracy and response time data. Speeded performance is required for computing scores	Ghana, Côte d'Ivoire, Argentina, Niger, Kenya, Brazil

CONSTRUCT	PARADIGM DESCRIPTION	ADMINISTRATION PROCEDURE	EXEMPLAR STUDY SETTINGS
Working Memory Non-Verbal	**Corsi Blocks/Dot Matrix** Children are presented with a spatial array of blocks or objects and are instructed to remember the sequence of selected blocks or objects by storing the information in their working memory (e.g., replicating the same or reverse order in which blocks/locations were selected). Children must be familiar with sequencing and, for backward items, doing things in reverse.	**Common Stimuli:** grids with blocks lighting up, spatial location of stickers on a cartoon ant/peanut, lights, cubes **Versions:** manual and digital tablet-based versions are used **Metrics:** digital tablet-based yields accuracy and response time data	Ghana, Côte d'Ivoire, South Africa, Ethiopia, Hungary, Argentina, Chile, Vanuatu, Malaysia, Brazil, Madagascar
Working Memory Verbal	**Digit Span/ Word Span** Children are instructed to repeat a sequence of digits, words, or sentences in the same and/or reverse order as what is presented to them by holding the order of the stimuli in their working memory and reciting the sequence. Children must be familiar with knowledge of the digits or words used as stimuli, and, for backward items, saying things in reverse.	**Common stimuli:** digits, words, sentences **Versions:** manual or digital tablet -based version **Metrics:** accuracy only for manual and response time available for digital	Nigeria, Chile, Cambodia, South Africa, Ghana, Madagascar, Pakistan, United States, China, Uganda
Cognitive Flexibility Non-Verbal	**Dimensional Change Card Sort (DCCS)** Children are presented with three different blocks of trials. In the first block, children are instructed to sort cards depicting colors and shapes according to one dimension (e.g., color). In the second block, children are asked to switch and sort the same cards by the other dimension (e.g., shape). In a third block of the task, children are instructed to flexibly switch between the two dimensions following an additional rule for when to sort by each dimension. There is an easier version of this task that separates two dimensions by using cards that depict black silhouettes of shapes against a colored background. This version could be used if the task with integrated color-shape stimuli is too challenging.	**Common stimuli:** playing cards with geometric figures, animals or everyday objects (truck, boat) **Versions:** manual or digital tablet-based tasks are both used **Metrics:** accuracy is the main metric used	South Africa, Pakistan, Kenya, Chile, Cambodia, Rwanda
Cognitive Flexibility Non-Verbal	**Flexible Item Selection Task (FIST)** Children are presented with three stimuli depicting figures that vary across three dimensions (color, size, shape) each with 3 values (e.g., for size, small, medium, and large). Children are asked to identify 2 different sets of matching stimuli. Children must flexibly switch between different dimensions when identifying a pair of stimuli that depict figures that are the same across one dimension (size value) and then a pair of stimuli that are the same on a different dimension (shape value). Children must be familiar with categorizing based on conceptual features. The GEFI network has created a new version of this task that adds more dimensions/values and stimuli cards and thus can be used with a wider age range. See the **GEFI website** for more info.	**Common stimuli:** geometric or everyday figures **Versions:** manual or digital tablet -based versions **Metrics:** accuracy is the main metric used	Kenya, Chile, Argentina, Rwanda

DEVELOPMENTAL CONSIDERATIONS

Younger children (ages 3-5) may not have the general cognitive skills (e.g., language) and knowledge (e.g., numbers, colors) required to successfully complete some EF tasks. Regardless of age, children who have limited or no experience with formal schooling (including exposure to early childhood care and education programs) may also struggle to complete tasks that rely on skills, knowledge, and experiences that are socialized in educational settings (e.g., adults asking hypothetical questions, back-and-forth interactions). Throughout this document, we highlight adaptations that researchers can use to minimize the impact of other skills and knowledge on children's performance on EF tasks.

By selecting tasks that require non-verbal responses (e.g., pointing, tapping, sorting) rather than verbal responses (e.g., saying a word), researchers can minimize confounding assessment of EF skills with children's verbal ability, as well as their comfort with speaking in front of an unfamiliar adult.





By selecting tasks that assess visual working memory, researchers can minimize confounding assessment of EF skills with children's knowledge of numbers and numerical symbols, as well as their comfort with speaking in front of an unfamiliar adult.

Photo Credit: South Africa, Project led by Cathi Draper (Above)
Pakistan, Project led by Dr. Aisha Yousafzai (Left)

CULTURAL CONSIDERATIONS

Most widely used EF tasks assess children's EF skills in a decontextualized way, following paradigms that do not resemble the everyday experiences and activities of children around the world. EF tasks are intentionally designed to be de-contextualized to minimize the effect of everyday non-EF factors (e.g., practice, expertise, interest). Further, existing tasks rely on certain behavioral engagement assumptions that are not universal across different settings and cultures and may produce unintended inequities in assessment experience and performance data. Thus, it is critical that EF tasks and procedures are adapted to be culturally appropriate to yield valid data (i.e., performance on a task captures individual differences in children's EF skills). To accomplish this, one should consider common features of standardized direct assessments that apply to administration of many performance-based EF tasks.

1 INTERPERSONAL INTERACTION & RAPPORT BUILDING

Are children comfortable and/or willing to interact with unfamiliar adults in one-on-one settings? Are children comfortable and/or used to being questioned by adults, and making direct eye contact with adults?

2 ACADEMIC/ABSTRACT THINKING

Are children familiar with abstract concepts and figures, two-dimensional representations of objects, and categories (e.g., colors, numbers, and size)? Are children willing to engage in abstract, decontextualized thinking activities such as remembering and mentally manipulating symbols (e.g., digits)?

3 PERFORMANCE RESPONSE

Are children comfortable and willing to perform in a way that could be seen as silly, inappropriate, or the opposite of what is perceived as the norm (e.g., saying night when shown a picture of the sun)?

4 PERFORMANCE MOTIVATION

Are children comfortable and willing to engage with game-like and repetitive tasks that do not have a clear everyday purpose? Are children comfortable and willing to ask questions when they don't understand instructions issued by an unfamiliar adult? What does compliance without comprehension look like?

5 TIME PRESSURE

Are children comfortable and willing to participate in timed tasks that prioritize a speedy response? Do children focus on accuracy over speed in such tasks?

These questions should be addressed in relation to cultural and contextual norms and used to guide the task selection as well as the adaptation of task instructions, administration procedures, and examination of collected data.

CULTURAL & CONTEXTUAL ADAPTATIONS

While the adaptation of task materials and administrative procedures will vary across cultures and contexts, we provide examples of how researchers adapted existing EF paradigms for administration with young children in rural Pakistan to make them more culturally meaningful, which in turn increased the comprehension and validity of collected data.

EXAMPLES FROM THE STUDY OF PRESCHOOLERS LIVING IN RURAL PAKISTAN (PIs: Aisha Yousafzai; Jelena Obradović)

Interpersonal Interaction and Rapport Building	Assessors were trained to sit on the floor with a child and engage in turn-taking play by rolling the ball back and forth with that child. This activity helped introduce children to the turn-taking nature of the assessment. Assessors then started the EF assessment with a simple, short task that required only pointing.
Abstract Thinking	A go/no-go task was adapted to include an image of a cat as a go-stimulus because children approach and pet stray cats, and an image of a dog as a no-go stimulus because children are told not to approach and pet stray dogs that can bite them. Young children are not familiar with words for colors, but were able to distinguish between "grass-colored" and "mango-colored" images to sort cards by color. Thus, these labels were adopted instead. The task included pre-task trials that tested children's understanding of these terms.
Performance Response	Young children were not familiar with puppets and did not want to perform actions issued by an assessor-controlled puppet, so a task that relied on images was used. Children also struggled saying an opposite term when presented with a card showing an abstract stimulus (e.g., say day when presented with a picture of night). However, they were able to state an opposite when the card depicted a physical characteristic of a familiar animal (e.g., say big when presented with a picture of a little cat).
Performance Motivation	Assessments were conducted by local female staff who were trained to make both child and parent comfortable, address the child's needs (e.g., hunger, nap), encourage participation, remain non-judgmental, and prevent the parent from interfering with the child's responses. The quality of performance data was further evaluated for perseverative responses (i.e., repeatedly selecting the same response) to investigate children's comprehension of task rules.
Time Pressure	Due to the very young age of participants, only untimed "table-top" tasks with tangible materials were used.

TASK VALIDITY

When considering administration and adaptation of existing tasks, it is critical that the validity of the selected EF paradigm is maintained. To ensure that the task truly measures the intended EF skill, interrogate whether task-specific paradigm assumptions are appropriately maintained in a study culture or context. We provide an example of how delay-of-gratification tasks have been misused, yielding invalid data and inaccurate conclusions in many cultures.

Delay of Gratification Example

Delay of Gratification can be conceptualized as a form of inhibitory control, in contexts where instant gratification is a dominant, prepotent response. Tasks can measure how well children delay gratification, or forego it entirely, usually to obtain later rewards. One common task requires a child to delay eating a small treat (e.g., candy) placed before them. In some versions of this task (e.g., the marshmallow task), if the child suppresses the impulse to grab and eat the treat, instead activating a subdominant response to delay eating for a certain amount of time, they will receive an additional treat. In cultures where children are socialized to wait and never grab food on their own, or to wait to commence eating until everyone is served, this EF task has repeatedly failed to yield valid EF data (i.e., everyone waits for the treat and there is no variation in children's performance, or the variation in children's performance reflects sensitivity to cultural norms and not use of EFs). This result should not be surprising, as the critical assumption of this task paradigm was not valid in these cultures. However, a study in Japan, where children are socialized to open gifts immediately, revealed that a task that focused on waiting to open a gift produced a valid measure of delay-of-gratification skills in young children. Dr. Yuko Munakata further explains **this work here: bit.ly/4dD1sC6** .

TABLET-BASED ASSESSMENT CONSIDERATIONS

Given widespread use of computerized (and specifically tablet-based) EF tasks in many global contexts, we highlight some key characteristics of this assessment modality to consider:

1

STANDARDIZED PROTOCOL & AUTOMATIC DATA RECORDING

Computerized EF tasks provide a more standardized assessment procedure, which eliminates the need for recording data by hand, and reduces the assessor training required to reliably administer EF tasks.

2

SCALABILITY AND FLEXIBILITY

Computerized tasks are particularly useful when studying a large sample or employing a large number of assessors and in multilingual contexts where the standardized instructions can be recorded and played in various languages.

3

GROUP ASSESSMENTS

Use of tablet computers has enabled researchers to conduct EF assessments with groups of children. Assessment in a group setting may more closely resemble children's everyday experience and thus increase ease and comfort with the assessment procedure. This approach may not be feasible with very young children who may need direct individual support to get through the assessment. With children's age and their increasing ability to independently sustain attention and engage with a tablet task, the size of the group can increase from a few children to an entire classroom.

4

RESPONSE TIME DATA

Use of computerized tasks also enables collection of response time data that can be used to measure EF skills in older children (typically ages 8-12) who may perform at ceiling levels on non-computerized tasks that only yield response accuracy data.



Photo Credit: Cote d'Ivoire, Dr. Michael Sulik

WI-FI REQUIREMENT

While computerized tasks that require Wi-Fi or mobile connectivity may not be feasible for use in remote areas where the Internet is not stable, there are several programs and platforms that enable offline data collection with delayed data upload when Wi-Fi is accessible.

FAMILIARITY WITH COMPUTERS

There are generally no issues with using tablet-based EF tasks with children who have little to no experience using a tablet, provided that tasks require only simple responses (e.g., press a button) and comprehension and engagement are appropriately scaffolded by an attentive assessor. Indeed, tablet-based administration may increase task compliance and motivation in places where children do not routinely interact with tablets, because they are perceived as novel and fun. However, if some children are reluctant to use tablets (e.g., after practice trials to determine comfort level), using non-computerized tasks that include tangible materials may improve children's engagement and performance. Regardless of children's comfort engaging with tablet-based EF tasks, ecological validity of tablet-based assessments will be limited in contexts where children do not have any experience seeing or using such devices and this should be taken in consideration when interpreting data.



Photo Credit: Bangladesh, Ishita Ahmed

TRANSLATION & CULTURAL ADAPTATION

The first step during the task adaptation phase is to conduct a high-level review of selected EF tasks with adults who have direct in-country and local cultural experience working with young children (caregivers, educators, health workers, researchers) and identify potential issues that should inform translation and iterative pilot testing (e.g., conducting a small-scale study using selected measures).

Next, researchers should use the following recommendations to guide their translation and adaptations of task materials and instructions.

TASK LANGUAGE

Identify in-country partners familiar with local dialects and language conventions to translate the task instructions and the accompanying script for assessments.

Provide translators with a high-level understanding of the objectives of the task. Traditional forward and backward translation processes may not capture nuances of EF tasks, especially for "child-friendly" instructions that may not be grammatically correct.

If words are used as task stimuli (e.g., word span task), ensure that selected words have equivalent difficulty across all translations (e.g., number of syllables, pronunciation, familiarity).

In communities where more than one language is commonly used, decide if it is feasible and appropriate to exclusively administer in the country's official language (especially if conducting research in an educational setting where that language is employed). If not, identify all languages that the tasks should be administered in and clarify procedures that assessors will use to determine the child's preferred language. Options include:

- Ask an adult who knows the child well (e.g., parent/caregiver/teacher) what language the child is most comfortable with
- Ask a child for their preference (although note that some children may feel the pressure to select the official language)
- Have the assessor converse with a child and decide based on the child's responses
- Use administrative data, such as school records to determine the child's primary language (although note that administrative data may be inaccurate or outdated)
- Conduct a formal language assessment

Create procedures describing what to do if assessments cannot be conducted in a child's preferred language and how this may affect analyses of that child's performance data.



TASK CONCEPTS

Some concepts and terminology may be culturally appropriate, but may not be developmentally appropriate (e.g., where adults have terms that name colors, but young children have not yet been taught words for colors). It is critical that task components are both culturally and developmentally appropriate.

Examine children's familiarity with the translated task in terms of:

- Task images that represent the stimuli (e.g., children may not recognize non-native animals, an image of a boat should resemble familiar types of boats);
- Language that labels the stimuli (e.g., "chicken" may be more familiar than "bird");
- Language that describes the stimuli (e.g., colors, size);
- Task components (e.g., in some contexts talking puppets may be unrealistic and/or confusing); and
- Language that communicates task concepts and/or instructions (e.g., requests to do the "opposite" or to repeat received information "backward").

If any pictures, words, phrasing, or concepts are unfamiliar to children, identify and pilot test local alternatives (e.g., in some African languages, the words for blue and green are the same, so using phrases "blue like the sky" or "green like the grass" helps differentiate them).

If the task requires differentiating or sorting by color, make sure that colors can be differentiated by children who may have trouble seeing differences between colors. Use online tools such as a color blind simulator **(bit.ly/3XYEArv)** to test whether task colors have enough contrast to be seen as separate by children with color vision deficiencies.

Include pre-task questions without EF demands to test if your instructions are working (e.g., test children's knowledge of color terms before practicing and testing sorting by color). Some tasks may already include these types of questions.

Note, some task stimuli include abstract patterns that by their design are unfamiliar to all children, so participants cannot use labels to simplify the task demands. Whenever possible, use culturally relevant examples such as patterns used in weaving, textiles, etc.

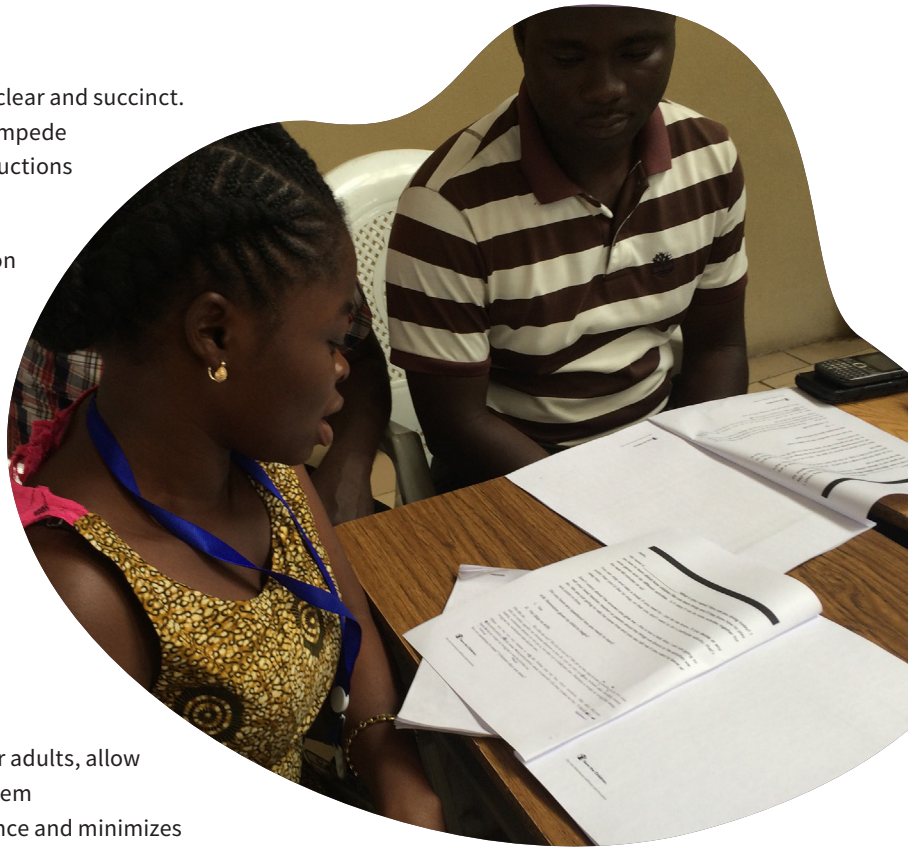
Photo Credit: Indonesia, Dr. Tony Okely

TASK ADMINISTRATION PROCEDURES

As needed, modify task instructions so that they are clear and succinct. Overly complicated vocabulary or explanations can impede understanding of task instructions. Long verbal instructions are also not common in many cultures.

Whenever possible, include non-verbal demonstration of the task instructions and provide options for non-verbal responses (e.g., pointing), especially in contexts where verbal responses to adult inquiry are not common for young children.

Engage local child development experts in explicit discussions of what would be the most appropriate way to get children comfortable with the assessment staff, space, and procedures and how to make the entire experience more culturally familiar. Participants may feel more comfortable if they have an opportunity to see or interact with the assessment staff prior to the start of the assessment.



If children are not used to being alone with unfamiliar adults, allow their caregivers to attend the session, but position them in a way that limits their view of the child's performance and minimizes potential interference with the assessment (e.g., some caregivers may be inclined to coax their child, provide verbal or non-verbal feedback, or answer on a child's behalf).

If children are not used to being directly questioned by adults in a one-to-one setting, have the assessor sit next to the child (instead of across) and minimize direct eye contact.

Inquire how common or unusual it is for adults to "play" with children and, if it is appropriate, engage in these types of activities. If culturally appropriate, using dolls or puppets can help young children understand instructions and may make them feel more comfortable when giving responses.

We encourage further examination and adaptation of the task materials, instructions and procedures based on the results of pilot testing.

Photo Credit: Ghana, Project led by Dr. Sharon Wolf

SELECTING TASK PARAMETERS

Task parameters should be calibrated to produce good variability in the final performance index/indices, which may include accuracy scores, response times, or their combination. Pilot testing can help researchers check task parameters that are appropriate to the age and skill level ranges of the study participants (see Piloting Tasks below).

The Number of Practice Trials

Practice trials are critical to ensuring that children understand all elements of the task and what they are expected to do. Children who are younger and also less familiar with standardized assessments or specific task demands will need more opportunities to practice each task rule. Consider adding an additional pre-practice trial that tests knowledge of non-EF task elements (e.g., colors, shapes, numbers).

The Number of Test Trials

While more test items will generally increase the reliability of EF performance data, the number of test items needs to be balanced with the length of the task and children's capacity to stay focused and engaged. Start with well-established conventions and adjust according to these principles as needed.

- Tasks designed to assess **inhibitory control skills** tend to have two types of trials (sometimes separated into different test blocks). For example, a go/no-go task consists of frequent trials with "go" stimuli to which children need to respond (by pressing a button) in order to build up a prepotent response and less frequent trials with "no-go" stimuli to which children need to not respond (by not pressing a button). Thus, there needs to be enough go trials that produce a prepotent response for this task to yield reliable data on no-go trials.

- Tasks designed to assess **working memory span** (e.g., dot matrix, digit or word spans) typically start easy and increase in difficulty. For a given difficulty level (e.g., 2-span trials, 3-span trials), consider how many trials to include. More trials will increase reliability but will also make the task longer. To reduce frustration for participants and task duration, you can implement a stopping rule after a specified number of incorrect responses in a row.

Task Intervals

Timing of different trial components varies across tasks and can influence task difficulty and children's performance. Some tasks are programmed to let researchers change these timing parameters.

> **Stimulus duration/response interval:** How long an image/trial is displayed and how long children have to respond to a trial by indicating their response.
>
> **Inter-stimulus interval:** This is the length of time between stimulus presentations.
>
> **Combination of stimulus duration and inter-stimulus interval** will place differential demands on processing speed. For example, the association between stimulus presentation time and task difficulty can vary depending on the type of task. Longer stimulus durations make inhibitory or cognitive flexibility tasks easier because they give children more time to respond and thus increase accuracy. In contrast, longer inter-stimulus intervals for working memory span tasks make those tasks harder because children need to hold the presented information in memory for more time before responding.

You might also adjust these intervals if a significant portion of children are not able to respond in time, which could be due to their age, familiarity with speeded responses, and/or cultural variation of children's level of engagement (see Piloting Tasks below).

TRAINING ASSESSORS

Employ assessors who speak the relevant local language(s), based on participants' language needs and preferences, and ideally who live in or are familiar with the community.

Train local assessors on best practices of task administration, which consists of being able to:

UNDERSTAND BASIC PRINCIPLES OF CHILD ASSESSMENT

Start by explaining why EF skills are being assessed and the intent of each task. Next, explain that the role of the assessors is to ensure that participants understand task requirements during instruction and practice trials, but not to interfere with performance during test trials. These principles might not always be familiar to assessors who might be more familiar with scaffolding a child toward a correct response (e.g., local teachers, care providers).

ESTABLISH RAPPORT WITH CHILDREN

Children who are not familiar with direct assessment or similar activities may need more time to get comfortable with the assessment situation and the unfamiliar research team. Use culturally relevant and developmentally appropriate activities to establish rapport. Discuss with the local assessors how much time is needed to build good rapport and what are observable indices that the child is comfortable (e.g., smiling may not indicate comfort across all cultures).

ADMINISTER A STANDARDIZED SCRIPT

Assessors should practice the script so that the assessment experience feels natural and more comfortable to children. The assessment protocol should explain how to troubleshoot common issues, such as what to do and what to say if the child is having difficulty understanding the script, to minimize assessors' need to improvise.



Photo Credit: Pakistan, Project led by Dr. Aisha Yousafzai

USE CULTURALLY-APPROPRIATE POSITIVE/AFFIRMING TONE OF VOICE AND FACIAL GESTURES

Do this for all children and their families. Do not imply an answer is correct or incorrect, but focus on their effort to complete the task ("I can see you are working hard, that's great!" or "Keep going, we are about halfway through already!"). Don't engage in verbal or non-verbal disapproval of a child's responses to the task, or of a caregiver's behavior.

RESPOND IN A PRE-PLANNED, STANDARDIZED WAY TO ANY DEVIATION FROM A STANDARDIZED PROTOCOL

Examples include: assessment disruptions or interruptions (e.g., outside noise, someone walks in, a child needs a bathroom break); a distracted child; a child who does not understand or hear instructions; a child who wants to stop; and a talkative child who is significantly slowing down the assessment.

RECORD OBSERVATIONS OF UNUSUAL ENVIRONMENTAL CONDITIONS DURING TESTING

Record assessment disruptions and complete a subjective rating of data quality based on conditions in the testing environment and child task engagement.

TRAINING & CERTIFICATION

Procedures should include (depending on available time and resources):

Reading and discussing the script as a group to understand the tasks and assessment procedures.

Watching recorded or live demonstration of the assessment protocol.

Assessors memorizing the script and procedures by practicing on their own and then in pairs or small groups.

Assessors demonstrating their knowledge of the script/procedures, including how to handle common problems and being certified by experienced research personnel. Assessors can be tested by doing a practice assessment with an experienced member of the research team playing the role of the child and purposefully introducing common problems. Having a certification checklist helps with this process. See page 21 for an example of an assessor certification checklist. Note that examples of what is an appropriate assessor response to children's off-task behaviors should be evaluated in a specific study context.

Observing assessors conducting practice assessments with children who are similar to the target population for the study, and providing feedback from the observing member of the research team about assessor performance. If the pool of participants is limited (e.g., children with developmental delay), practice can be done with an alternative population with similar skill levels (e.g., younger children without a diagnosed developmental delay).

Live and/or recorded observation (e.g., video on a mobile phone) and review of the assessor's first assessment(s) in the field.

Data quality

Data quality should be monitored on an ongoing basis and assessors should receive feedback on their performance by research team leaders. This may include a systematic review after a percentage of assessments is complete (e.g. after five kids; then, if going well, after 10 kids; then 20, etc). As data is collected, test whether there are systematic differences in child performance or time taken to administer tasks across assessors (record the name of assessor on each data record). If needed, address differences through additional training. Using an assessor certification checklist (see example on page 21) can help reduce these assessor effects.

ASSESSOR CERTIFICATION CHECKLIST EXAMPLE



- ◯ Engages the child by presenting each game with a lively and positive voice
- ◯ Knows and follows the script for each game/task
- ◯ Ensures the child has clearly understood task requirements before they proceed
- ◯ Remains engaged throughout the assessment
- ◯ Provides generic positive feedback to maintain the child's engagement
- ◯ Responds appropriately when the child wants to stop (e.g., gently encourage persistence by praising effort and highlighting progress, stop if encouragement fails)
- ◯ Responds appropriately when the child is too talkative (e.g., acknowledge interest to share and connect, but redirect to the task by emphasizing its importance)
- ◯ Responds appropriately when the child is shy/ does not engage with task (e.g., take time to build a rapport, praise effort)
- ◯ Responds appropriately when the child is not paying attention (e.g., remove distractions, increase engagement by highlighting fun aspects of the task)
- ◯ Can stay calm if the child does not understand the tasks
- ◯ Doesn't give feedback to the child about their performance during the task
- ◯ Doesn't provide hints or scaffolding to support a child's performance/responses
- ◯ Accurately documents any deviations from the protocol and reasons for missing data (e.g., why a child stopped responding)
- ◯ Accurately completes the assessor-report survey of child's behaviors based on the child's behavior during the session

ASSESSMENT CONSIDERATIONS

ASSESSMENT SPACE

Conventionally, EF assessments are conducted in a setting with minimal environmental distractions so the child can remain focused on the task. The setting can be indoors or outdoors. If a caregiver is present (a recommended practice when assessing young children who may have separation anxiety), instruct them how not to interfere with or distract from the child's task performance. If other adults and children are nearby (e.g., home setting, school playground), the assessment should take place in a separate location or in a corner of the shared space where the assessor can try to minimize distractions and interference (e.g., siblings trying to watch or engage in the assessment tasks). Ensure that the administration/testing context matches a child's typical or culturally-appropriate experience to promote the child's comfort level and engagement. For example, in many cultures it is more appropriate for an assessor to sit next to a child than across from them. This seating arrangement also minimizes direct eye-contact with an assessor, which is important in cultures where children have limited or no face-to-face, one-to-one interaction with unfamiliar adults.

NOISE/DISTRACTIONS

Typical background noise is less concerning than unexpected noise that draws the respondent child's attention away from the assessment. Consider what would make the child most comfortable in the assessment space (e.g., for children who are not familiar with quiet spaces, being in a quiet space may make them uncomfortable and may negatively affect their performance). It is important to balance the need to minimize distraction with the child's comfort level and ability to remain focused on the task. Events that cause disruptions in assessment procedures (e.g., staff/family entering or exiting the assessment space, off-topic conversations within a child's earshot) should be minimized.

> Consider training assessors to rate (a) **the level of noise** (1: no external noise; 2: low-level, background noise; 3: moderate/high noise), and (b) **the degree of disruptions** (1: no unexpected/abrupt disruptions, 2: a couple of unexpected/abrupt disruptions; 3: repeated unexpected/abrupt disruptions). These ratings can be used to examine or control for any systematic differences in children's performance due to noise/distractions.

CHILD-LEVEL CONFOUNDS

Consider collecting the following data to interrogate known child-level EF covariates:

1. Age
2. Gender
3. Years of schooling (including pre-primary and early childhood care centers)
4. Language of assessment
5. Language background (e.g., preferred language, primary language at home, self-reported proficiency in other languages) and/or language ability (e.g., performance on a standardized vocabulary assessment in the assessment language)
6. Socio-economic status or related proxies (e.g., parent's education level)
7. Community characteristics (e.g., urban/rural)
8. Assessor notes regarding assessment context and quality (e.g., significant disruption or disengagement)

There is a growing expectation that these are captured and reported to contextualize findings.

ASSESSOR-LEVEL CONFOUNDS

Consider collecting these data about the assessor to identify potential sources of assessor bias:

1. Relevant demographic characteristics (e.g., gender, ethnicity, socio-economic background)
2. Familiarity with the child (e.g., parent, teacher, stranger)
3. Beliefs/mindsets that may impact assessor's behaviors when interacting with children (e.g., class or race stereotypes and biases).

This data can be used to understand and interpret your EF data.



Photo Credit: Côte d'Ivoire, Dr. Michael Sulik

PILOTING TASKS

It is important to pilot test the selected tasks with a small number of children who are similar in age and background to those who will be included in the main study to understand the validity of the tasks for the specific study. Piloting should be done even if selecting tasks that have been used with children of similar age and in similar contexts. Pilot testing should address whether:

1. Children understand, actively engage with, and complete the tasks
2. Assessors know how to appropriately administer the tasks
3. The tasks capture sufficient variability in children's performance

Start with 5-10 children and expand the pilot sample if necessary to resolve any issues that come up (e.g., multiple children struggle with the task and more data is needed to understand what revisions are needed). If feasible, pilot assessments should be video recorded so that experts who are providing feedback do not need to attend the assessment session (i.e., having additional people physically present in the assessment space may contribute to the child's discomfort and affect their task engagement). Recording can also be used for training purposes.

CHILD ASSESSMENT EXPERIENCE

Engage local assessors in a group debriefing by discussing children's observed engagement and performance and what is and is not working when administering each task. Revise assessment instructions or approach if there are consistent problems across multiple administrations and there is a consensus across multiple assessors. It is important to avoid a single assessor's impression having undue influence on a task or making changes if only one or two children are not able to complete the task. It is expected that some children, and especially younger ones, will find some tasks challenging. If possible, pilot assessments should be observed and analyzed by someone who has experience using the same tasks in other contexts because this person can provide an important point of comparison about how well the tasks are working (e.g., understanding the instructions, performance, and level of engagement) and what is not working. Lastly, if developmentally appropriate, interview children to understand their assessment experience and responses.

ASSESSOR PERFORMANCE

A piloting phase should also be used to certify assessors who are ready to conduct the study by observing their administration skills (see page 21 for a checklist example). This is a great time to clarify any procedures for when the child does not comply or understand task instructions, is distracted or uncomfortable, or needs a break. Administering the entire protocol during the pilot phase will identify any operational challenges in administering all assessments in a given time and space. If administering the entire protocol is not feasible, assessments can be segmented into separate sessions and assessors trained to address any fatigue and transition issues.

DATA VARIABILITY

Examine pilot data to identify if administered tasks are yielding adequate variability in children's performance. Specifically, identify how many children are:

1. Passing comprehension criteria
2. Hitting floor and ceiling
3. Perseverating by choosing the same response
4. Not engaging with the task or not completing the task.

Examine how these performance metrics vary by age and any other relevant demographic characteristics. If pilot data reveals issues with a speeded task (e.g., most kids are not responding within allotted time limit on most trials), we recommend pilot testing different stimulus presentation/response times to determine what is most appropriate for your sample (e.g., for producing good variability without floor or ceiling effects).

Pending available time and resources, conduct additional rounds of piloting, making sure both tasks and procedures are developmentally and culturally appropriate (as judged by pertinent community members in that culture), and capture significant variability in children's performance. If pilot data reveal persistent issues with engagement, comprehension, and/or data quality, re-consider your initial task selections, adaptations, and/or task parameters.



Photo Credit: Papua New Guinea, Dr. Tony Okely

DATA PROCESSING & CREATING EF VARIABLES

For each task, determine if performance will be based on trial-level accuracy, trial-level response time (RT), and/or combination of accuracy and RT. This distinction is particularly relevant for speeded tasks (attention, inhibitory control), in which there is often a developmental shift with age (e.g., researchers tend to rely on accuracy only for younger kids and shift to accuracy + RT or RT exclusively in older kids).

The following procedures help with processing, evaluating, and aggregating collected EF data, but are often under-reported in published EF research.

ACCURACY DATA

LACK OF RESPONSE

If the child fails to respond to a test trial, determine whether this lack of response will be scored as: (1) missing data; or (2) incorrect response.

- For some tasks, there may already be consensus about how to score missing responses (e.g., for a no-go task, it is typical to score non-responses to "go" trials as incorrect).
- Where consensus is not clear, if the majority of children were able to respond in allotted time to most trials (i.e., response time interval was adequate), it is justifiable to score a missing response as incorrect. Note, this approach will have an effect on total accuracy score.
- If the child has not responded to many trials (especially consecutive trials), evaluate the assessor report for reasons why the child has failed to respond. If the child has disengaged from the task, the non-response trials should be coded as missing data as they are not a reflection of the child's skill and effort.
- If you score a lack of response as missing data, decide what minimum percentage of complete trials is needed for a task to be scored and included in analysis. This can be informed empirically by examining reliability statistics (e.g., what is the alpha reliability if the composite is restricted to responses to only four trials?).
- **Advanced recommendation**. You can also investigate the implications of this decision by creating two versions of the accuracy scores: (1) an accuracy composite that only includes trials with a response (i.e., non-responses are scored as missing); and (2) an accuracy composite that includes trials without a response, which are scored as incorrect.

PERSEVERATIVE RESPONSES

If participants press the same button repeatedly across many consecutive trials, it may indicate that they do not understand the task rules or do not want to engage with the task. Data analysis should address perseverative responses (e.g., by excluding those responses from accuracy calculations).

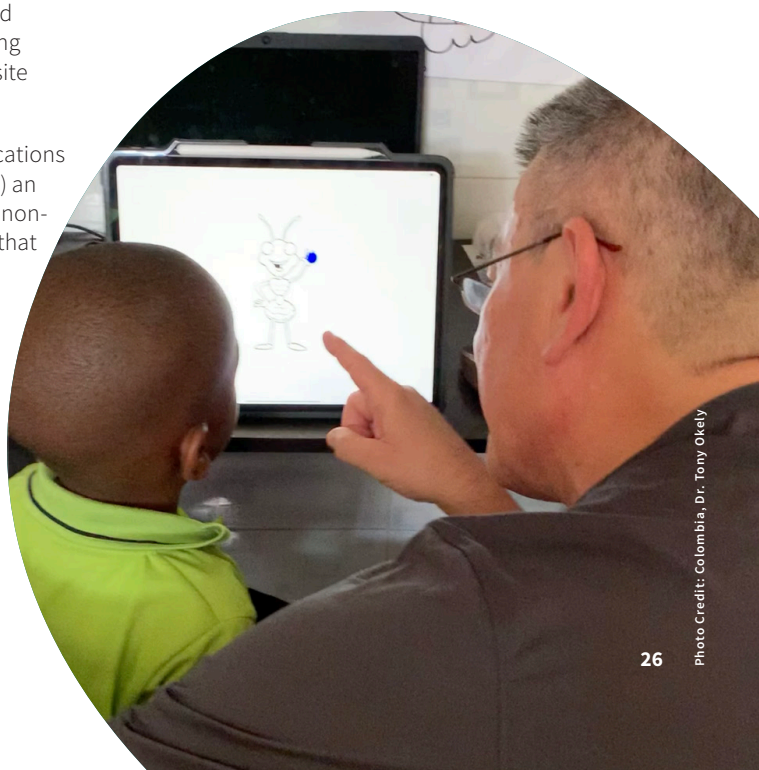


Photo Credit: Colombia, Dr. Tony Okely

ACCURACY DATA CONT.

ACCURACY INDEX

An aggregate score of accuracy data is commonly computed by averaging the binary-coded responses (0 = inaccurate, 1 = accurate) across all trials. In addition to aggregated accuracy, performance on working memory tasks can also be represented by the longest span the child can accurately complete.

DISTRIBUTION AND OUTLIERS

Examine the distribution of an aggregated score and check for outliers. Extremely low scores tend to be rare and can have a disproportionately large effect on analysis results unless recoded to be less extreme or excluded.

- If outliers are an accurate representation of the child's performance, they can be recoded to less extreme values (e.g., +/- 3 SD from the mean).
- If outliers do not represent valid responses (e.g., a child stopped responding because they lost interest in the game), they should be set to missing.



Photo Credit: Côte d'Ivoire, Dr. Michael Sulik

OVERLY RAPID RESPONSES

If the child responds within a fraction of a second (e.g., <300 milliseconds), their response may not reflect engagement of their higher-order cognitive processing (i.e., EF skill). The presumption is that these responses reflect a "trailing" decision/action from the previous item or impulsive, anticipatory response. Since such fast responses may not be a valid measure of a child's EF, trials with a response time <300 ms can be scored as missing for accuracy and response time or as incorrect.

RT INDEX

An aggregate score of RT data is computed by averaging trial-level RT data only for accurate trials. If the child's response is inaccurate, RT data is not a valid measure of cognitive effort (i.e., EF skills). Thus, it is critical to recode RT data to missing for all incorrect trials before calculating the RT index.

RELIABILITY OF RT INDEX

Examine the minimum number of accurate trials needed to have a reliable RT index by calculating reliability for different numbers of correct trials. Typically, at least 3-4 correct trials are required to have an acceptably reliable RT index (e.g., $\alpha > .7$).



Photo Credit: Kenya, Dr. Tony Okely

CONSOLIDATING MULTIPLE EF INDICES

If multiple EF tasks have been administered, aggregating the child's performance scores across all tasks can increase validity of EF assessment by covering multiple domains of EFs (i.e., inhibitory control, working memory, cognitive flexibility) or various aspects of a single domain (e.g., different IC tasks). Although EF tasks tend to have low-to-moderate intercorrelations, aggregating across multiple tasks can also increase the reliability of the EF composite.

- **If interested in studying specific types of EFs** (e.g., working memory vs. inhibitory control), one could aggregate EF performance across a subset of EF tasks that all measure a specific type of EF. Using a single task to represent one type of EF skill will confound task-specific influences on performance. If possible, use multiple tasks that assess a specific type of EF.

- **If an EF task has multiple blocks,** consider averaging scores within tasks (e.g.,average accuracy for the hearts and flowers and mixed blocks together) as scores within tasks tend to be more highly correlated than scores across tasks due to shared task-specific variance. This ensures that the overall EF composite does not overweight tasks with multiple blocks/scores. Exception: You may be specifically interested in examining differences across EF subdomains (e.g., inhibitory control vs. cognitive flexibility) that are assessed by the same task.

- **Aggregate accuracy scores.** Aggregating scores across multiple EF tasks is commonly done by averaging performance across multiple EF tasks after all task scores have been placed on a standardized metric (e.g. z scores, which reflect how many standard deviations the child is from the mean score in the sample) or transformed into percentile scores. It is also possible to use structural equation modeling to analyze EF performance across multiple tasks using a latent variable approach.

DATA ANALYSES, REPORTING, AND INTERPRETATION



Descriptive Statistics

Providing task-specific information advances scientific understanding of how different tasks work in different settings, even when the analysis variable is an aggregate score across multiple EF tasks. For each EF variable and EF composite, it is recommended to examine and report:

Task description and any adaptations: instruction, stimuli, number of trials, practice rules, stop rules, presentation times (both stimulus and inter-stimulus intervals), and metrics recorded.

Reliability for trial-level accuracy and response time data (if response time is intended as an EF indicator) e.g., using Cronbach's alpha based on correlations among all trials.

Data coding, transformations, and how composites were calculated.

Prevalence of and procedures for outliers and missing data.

Basic distributions and descriptive statistics on response variability (mean, standard deviation, range, floor & ceiling effects) for each EF variable. This helps other researchers understand the distribution of EF scores in the study sample and the average level of performance.

Photo Credit: Bangladesh, Ishita Ahmed

Associations

Examining how strongly EF variables are related to each other (within and across tasks) and to other key constructs advances the understanding of how EF assessments are working or not working and whether EF tasks capture meaningful variability in children's performance. To accomplish this, it is recommended to examine and report:

How descriptive variability statistics (range, floor/ceiling, any meaningful thresholds) vary by key child-level covariates and especially indices of developmental maturity (i.e., age).

How accuracy and RT variables are related and whether this association varies as a function of task, age, and accuracy level. Do not assume faster RT always indexes greater EF skills.

How performance on EF tasks relates to other assessments of cognitive skills or adult reports of EF-relevant behaviors in home or educational settings. These associations can provide valuable information on predictive and construct validity.

How performance on EF tasks is related to other child outcomes of interest to understand predictive and criterion validity.

When the raw scoring metric is meaningful (e.g., percent accurate), report unstandardized coefficients in addition to standardized coefficients to facilitate comparisons across studies.

Advanced recommendations. If interested in subgroup comparisons, establish measurement invariance across relevant demographic/culture/language groups, and/or time periods.

Examine effects of collected assessor/assessment variables: (1) relevant demographic characteristics (e.g., sex/gender, race/ethnicity, socio-economic background); and (2) beliefs/mindsets that may impact assessor's behaviors when interacting with children. If there is a significant assessor effect, consider clustering standard errors or including assessor fixed effects in models.

Interpretation

While EF skills have been conceptualized as universal skills that support goal-directed behaviors and have been shown to predict school readiness and academic achievement across diverse cultural settings, a given EF assessment is not necessarily equally relevant or equitable in capturing individual differences among children from varied cultural backgrounds. With this in mind, researchers and others who use EF tasks to assess children's skills should address the following considerations when interpreting their EF data.

Describe the sociocultural background of the studied population: (1) date of task administration in years (and, if relevant, months) and place of data collections (e.g., urban/peri-urban/rural, region, country); (2) the relevant sample characteristics (e.g., age in months if possible, race/ethnicity, socioeconomic status, languages, gender identity, religion, indigenous status, tribal membership, immigration generation). In addition, describe the participants' relevant socialization experiences (e.g., formal child care and schooling, family routines, home stimulation, developmental and social expectations, cultural values) as it pertains to the study's goals.

Discuss how EF skills are valued in this context and how selected EF tasks may or may not reflect children's everyday experiences. For example, consider children's familiarity with timed games and cultural values regarding speed/efficiency when interpreting RT data.

Since EF tasks measure individual differences among children, it is easy to forget that development of EF skills depends on contextual opportunities, incentives, and support. Group differences in children's performance on EF tasks will in part reflect differences in these broader contextual variables that reflect chronic structural inequities, underinvestment in disadvantaged communities, and pervasive experiences of discrimination. Misattributing and misinterpreting group differences in EF task performance as EF deficits only perpetuates harm inflicted on children and families coping with these inequities and injustices.

- Low EF scores should be interpreted in the context of systemic issues such as food security, housing stability, healthcare, child care, and education quality, as well as chronic experiences of stress, stigmatization, and/or discrimination.
- Children's performance on EF tasks will also reflect their familiarity and comfort with and engagement in the assessment tasks and context, as well as their acute state (e.g., fatigue, hunger, stress, illness, safety). These differential experiences will not only affect children's performance motivation, but will also make the tasks more or less challenging (see above for recommendations on how to probe this effect).
- If a significant proportion of children did not engage with the tasks, interpretation of EF data should focus on why tasks did not work rather than on children's skills.

Discuss how selected EF assessments reflect skills that are relevant for the key study outcomes and EF assessment limitations in capturing other relevant skills and competences. For example, performance on EF tasks relies on children's ability to maintain focused attention, which is also valued in formal educational settings. However, this type of attention may be contrary to certain types of learning approaches. "Open attention" or ability to be aware of and keep track of peripheral stimuli is more valued or adaptive in some contexts and cultures, and excelling at this type of attention may undermine one's performance on EF tasks. See **www.mdpi.com/2076-3425/14/4/318** for in-depth discussion of how cultural preferences may affect validity of EF data.

Discuss how studied contexts, outcomes, and/or experiences may privilege certain groups of children and how findings may offer directions for addressing these system-level biases. For example, some behaviors and competencies may be valued only in contexts where formal schooling is prevalent or among children growing up in nuclear families, urban settings, and Western cultures. Engaging with local cultural, sociological, or anthropological scholarship will advance understanding of how EF skills are valued, used, and promoted.

IMPLICATIONS

EF assessments have been designed and primarily used to study development of EF skills, including what promotes and hinders it, and to evaluate impacts of relevant practices, programs, and policies. However, global interest in EF skills has led to the use of EF assessments in educational settings that may require providing feedback to caregivers, communities, and other stakeholders. When considering the implications of EF data and related findings, one should:

Engage with local stakeholders to identify appropriate ways in which to contextualize and provide feedback on results, both in terms of individual feedback (e.g., to a child's caregivers) or average results (e.g., to a classroom teacher or school administrators). Research assessments have rarely been used to provide individual-level results. Use care when reporting individual-level results because they can be misunderstood or misused (e.g., supports could be misaligned to a child's actual progress and needs), particularly when they are based on a child's performance on one or two tasks.

Compile feedback in a way that is simple and easy to understand for a non-scientific audience using an infographic with metrics that can be meaningfully interpreted by a general audience. Generally, providing raw or standardized scores can be intimidating and/or difficult to understand for a non-scientific audience. However, until more EF data is available to establish meaningful population norms across diverse settings and cultures, we warn against sharing and interpreting individual children's scores.

Frame feedback within the context of the community's developmental goals and values related to children's performance (e.g., does the community/culture value EF skills and want to promote them in home and/or educational contexts?).

Present this feedback in a way that does not perpetuate a deficit perspective, while still considering how caregivers and stakeholders could promote improvement in EF (bearing in mind all considerations about cultural relevance mentioned above). While EF does develop with age, its developmental trajectory is highly variable within and between communities and contexts. So, formative feedback should focus on supporting a child's further growth, rather than emphasizing comparisons. For example, one can focus on how different environments can support or undermine the development of EF skills, including specific strategies such as ensuring adequate nutrition and sleep, providing culturally appropriate sensitive and responsive interactions with children, and strengthening culturally appropriate routines and experiences that enable children to practice EF skills. To avoid casting blame on adults in children's lives, address stressors and inequalities caregivers are facing and suggest changes to programs and policies that can support these practices at scale.

Practices, programs, and policies that aim to support a child and/or caregiving system may also be contextually bound in a similar way to assessment, so a well-calibrated EF assessment is also a precursor to research into what works, for whom, under what conditions across diverse contexts.

ADDITIONAL RESOURCES

Visit GEFI.Stanford.edu

Notable resources & readings:

Suggested readings
on EF-related topics

Parallel forms of parent and teacher surveys
of EF-related behaviors

References for studies
using recommended tasks in global settings

Assessor reported survey items
that can be used to rate self-regulation behaviors that the child displays while interacting with an assessor during a standardized assessment session

GEFI
Global Executive Function Initiative

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